

Mathematics Stage 4

Area of Circles and Sectors

Book 1 Develop and use the formula to find the area of circles and sectors to solve problems

Version: 261001

Feedback:
<https://MrDingMaths.com>

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Syllabus Content

MA4-ARE-C-01 applies knowledge of area and composite area involving triangles, quadrilaterals and circles to solve problems

Develop and use the formula to find the area of circles and sectors to solve problems

- Develop and apply the formula to find the area of a circle: $A = \pi r^2$, where r is the length of the radius
- Explain how the area of a sector can be developed from the area of a circle $A = \frac{\theta}{360} \times \pi r^2$
- Find the area of quadrants, semicircles and sectors, and apply these formulas in the context of real-life problems
- Calculate the areas of composite shapes involving quadrants, semicircles and sectors to solve problems

Area of a Circle

Review

☐ Square numbers

a 4^2

b 8^2

c 11^2

d 1.3^2

☐ Round decimals to 2 d.p.

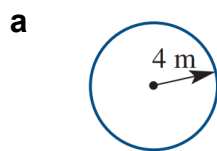
a 78.539845

b 153.938881

c 28.27454

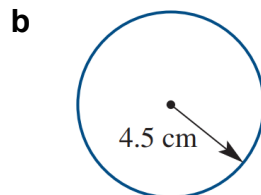
d 95.03399

☐ Identify radius and diameter of circles



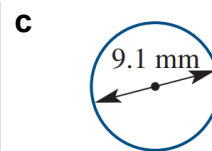
Radius:

Diameter:



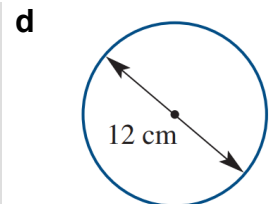
Radius:

Diameter:



Radius:

Diameter:



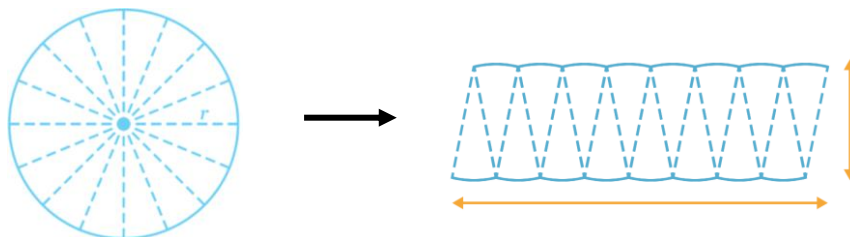
Radius:

Diameter:

? Investigation Area of a circle

Area of a Circle: <https://www.geogebra.org/m/fuwvtu7w>

Consider a circle divided into sectors, then placed together into a shape that resembles a parallelogram.



Label the base and height of the parallelogram, then calculate the area.

$$A = bh$$

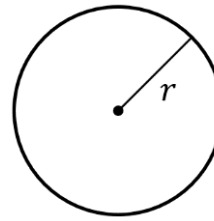
$$= \dots \times \dots$$

$$= \dots$$

i Area of a Circle

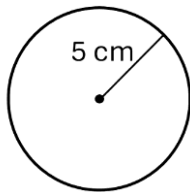
$$A = \pi r^2$$

↑
radius

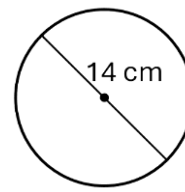


Example Calculate area of a circle

Calculate the area of these circles. Answer to 2 decimal places.



$$\begin{aligned} A &= \pi r^2 \\ &= \pi(5)^2 \\ &= 78.53981634 \dots \\ &\approx 78.54 \text{ cm}^2 \end{aligned}$$

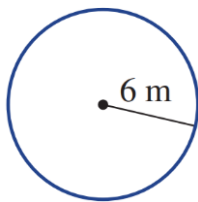


$$\begin{aligned} A &= \pi r^2 \\ &= \pi(7)^2 \\ &= 153.93804 \dots \\ &\approx 153.94 \text{ cm}^2 \end{aligned}$$

Guided Practice

Calculate the area of these circles. Answer to 2 decimal places.

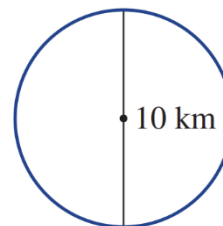
a



$$\begin{aligned} A &= \pi r^2 \\ &= \pi(\quad)^2 \\ &= \dots\dots\dots \end{aligned}$$

113.10 m²

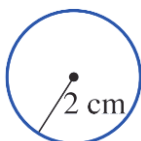
b



$$\begin{aligned} A &= \pi r^2 \\ &= \pi(\quad)^2 \\ &= \dots\dots\dots \end{aligned}$$

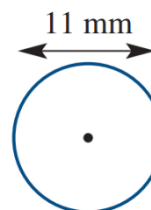
78.54 km²

c



12.57 cm²

d



95.03 mm²

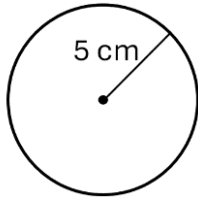
Key Ideas

1. We calculate the area of a circle using the formula
2. The formula always uses the of a circle
3. If the diameter is given, we need to change it to by the diameter before doing any calculations.

Interpret Exact form

Giving an exact answer in terms of π means we do not press the π button on the calculator. Instead, we write the number multiplied by π .

For example:



Exact form

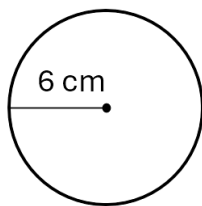
$$\begin{aligned} A &= \pi r^2 \\ &= \pi(5)^2 \\ &= 25\pi \end{aligned}$$

Rounded to 2 d.p.

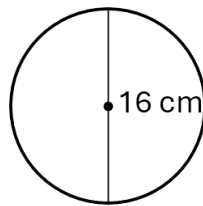
$$\begin{aligned} A &= \pi r^2 \\ &= \pi(5)^2 \\ &\approx 78.54 \text{ cm}^2 \end{aligned}$$

Find the exact area of these circles in terms of π .

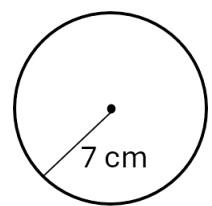
a



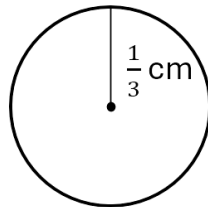
b



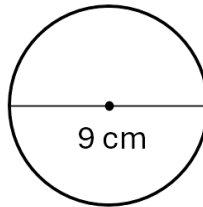
c



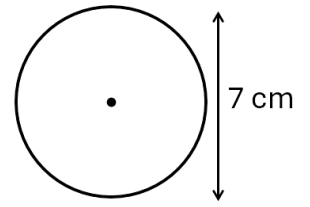
d



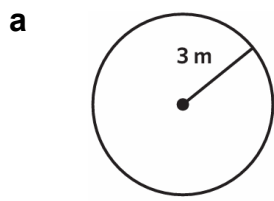
e



f

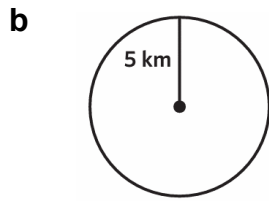


- 1 Calculate the area of these circles. Leave your answer in terms of π .
The first question has been done for you.

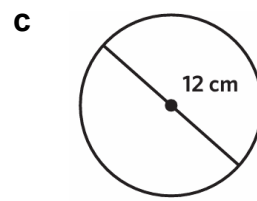


$$A = \pi(3)^2$$

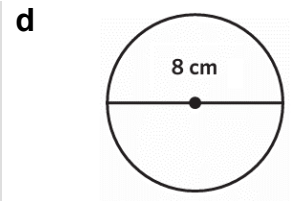
$$= 9\pi \text{ m}^2$$



$$25\pi \text{ km}^2$$

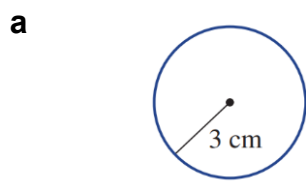


$$36\pi \text{ cm}^2$$



$$16\pi \text{ cm}^2$$

- 2 Calculate the area of these circles correct to two decimal places.

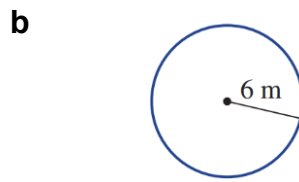


$$A = \pi r^2$$

$$= \pi(\quad)^2$$

$$= \dots\dots\dots$$

$$28.27 \text{ cm}^2$$



$$A = \pi r^2$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

$$113.10 \text{ m}^2$$

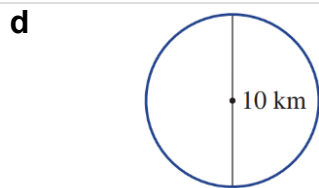


$$A = \pi r^2$$

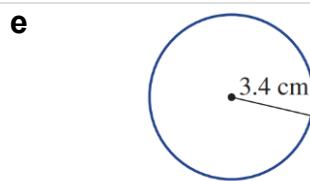
$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

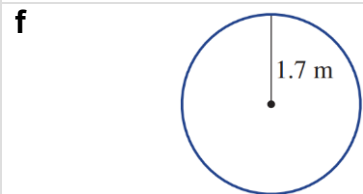
$$7.07 \text{ mm}^2$$



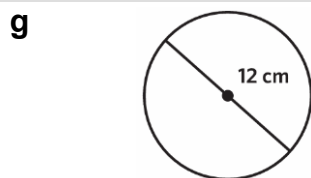
$$78.54 \text{ km}^2$$



$$36.32 \text{ cm}^2$$



$$9.08 \text{ m}^2$$



$$113.10 \text{ cm}^2$$



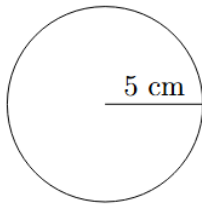
$$143.14 \text{ cm}^2$$



$$11.45 \text{ mm}^2$$

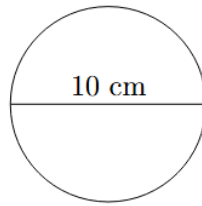
3 Calculate the area of these circles. Answer in exact form in terms of π .

a



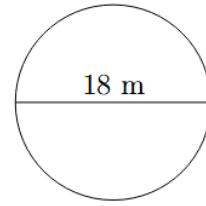
$$25\pi \text{ cm}^2$$

b



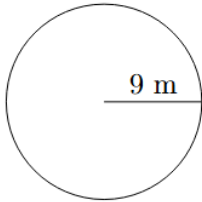
$$25\pi \text{ cm}^2$$

c



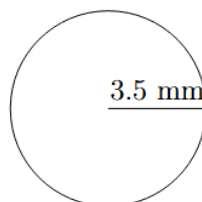
$$81\pi \text{ m}^2$$

d



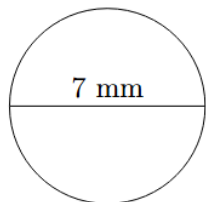
$$81\pi \text{ m}^2$$

e



$$\frac{49\pi}{4} \text{ mm}^2$$

f



$$\frac{49\pi}{4} \text{ mm}^2$$

4 A pizza tray has a diameter of 30 cm. Calculate its area to the nearest whole number of cm^2

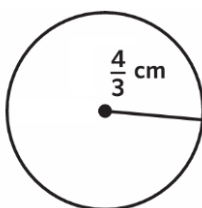
$$707 \text{ cm}^2$$

5 Two circular plates have radii 12 cm and 13 cm.
Find the difference in their area, correct to two decimal places.

$$78.54 \text{ cm}^2$$

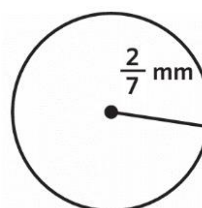
6 Find the area of the circle in terms of π as a fraction.

a



$$\frac{16}{9}\pi \text{ cm}^2$$

b



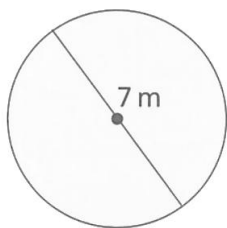
$$\frac{4}{49}\pi \text{ mm}^2$$

7 A tanker ship spills a circular oil slick 1 km in diameter.
The news reports that the area covered by the oil slick is over a square kilometre.
Are they correct?

No, they are incorrect. The area is only 0.785 km^2

- 8 Ed calculated the area of this circle as $49\pi \text{ m}^2$.

a Explain why he is incorrect.



The radius is $\frac{7}{2}$, not 7

b Write the correct area in terms of π as a fraction

$$\frac{49}{4}\pi \text{ m}^2$$

- 9 A farmer uses a sprinkler that waters a circular area. The sprinkler reaches 12 m in every direction. Calculate the area watered, to the nearest m^2 .

$$r = 12 \text{ m}$$

$$A = \pi \times 12^2 = 144\pi \approx 452 \text{ m}^2$$

- 10 A circular swimming pool has a diameter of 4.5 m. Calculate the area of the pool surface to 2 d.p.

$$r = 4.5 \div 2 = 2.25 \text{ m}$$

$$A = \pi \times 2.25^2 \approx 15.90 \text{ m}^2$$

- 11 A circular table has radius 60 cm. A tablecloth must overhang by 15 cm on all sides. What is the area of the tablecloth to the nearest cm^2 ? (Hint: find the radius of the tablecloth first.)

$$\text{Tablecloth radius} = 60 + 15 = 75 \text{ cm}$$

$$A = \pi \times 75^2 \approx 17\,671 \text{ cm}^2$$

- 12 A circular mirror has diameter 55 cm. A square mirror has side length 55 cm. The shopkeeper says the circular mirror has more glass. Is she correct? Show your working.

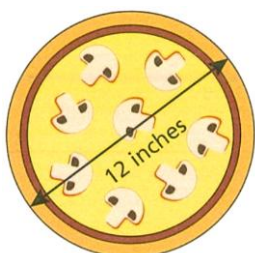
$$\text{Circular mirror: } A = \pi \times 27.5^2 \approx 2375.83 \text{ cm}^2$$

$$\text{Square mirror: } A = 55^2 = 3\,025 \text{ cm}^2$$

No, the square mirror has more glass.

- 13 Pizza sizes are given using the diameter, measured in inches. The 12-inch pizza is cut into 8 equal slices. Rayyan eats 3 slices.

What is the area of the pizza that Rayyan ate? Answer in square inches to 1 d.p.



$$\text{Whole pizza: } \pi(6)^2 = 36\pi \text{ in}^2$$

$$\frac{3}{8} \times 36\pi = \frac{27\pi}{2} \text{ in}^2 \approx 42.4 \text{ in}^2$$

- 14** Find the area of these circles with given radius. Give exact answers in terms of π .

$$r = 5 \text{ cm} \rightarrow A = \dots\dots\dots \text{ cm}^2$$

$$r = 10 \text{ cm} \rightarrow A = \dots\dots\dots \text{ cm}^2$$

$$r = 30 \text{ cm} \rightarrow A = \dots\dots\dots \text{ cm}^2$$

$$r = 5 \text{ cm} \rightarrow A = 25\pi \text{ cm}^2$$

$$r = 10 \text{ cm} \rightarrow A = 100\pi \text{ cm}^2$$

$$r = 30 \text{ cm} \rightarrow A = 900\pi \text{ cm}^2$$

What happens to the area when you double the radius?

The area multiplies by 4 (since $A = \pi r^2$, doubling r gives $\pi(2r)^2 = 4\pi r^2$).

What happens to the area when you triple the radius?

The area multiplies by 9 (since $\pi(3r)^2 = 9\pi r^2$).

- 15** You order a 9-inch pizza from a restaurant.

After a while, the waiter tells you that they no longer have 9-inch pizza bases and will instead give you two 5-inch pizzas instead.

He tries to convince you, saying that you will get more since $2 \times 5 = 10$.

Explain to him why this is not a good deal.

Note: pizza sizes are measured according to the diameter.

$$\text{9-inch pizza: } \pi(4.5)^2 \approx 63.62 \text{ in}^2$$

$$\text{2 5-inch pizzas: } \pi(2.5)^2 \times 2 \approx 39.27 \text{ in}^2$$

This is not a good deal as two 5-inch pizzas are significantly smaller than a single 9-inch pizza

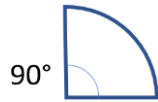
(This happens because the radius gets squared; if you halve the radius, the area is a quarter, so giving two pizzas won't make up the difference)

Area of a Sector

Review

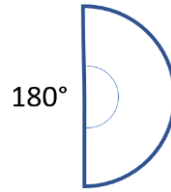
☐ Identify fraction of a circle shown

a

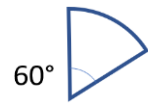


$$\frac{\square}{360} = \frac{\square}{\square}$$

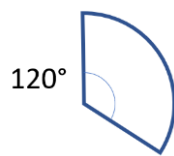
b



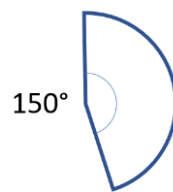
c



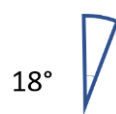
d



e



f

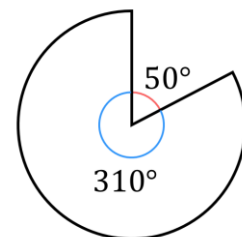


? Identify Interior angle of a sector

A sector is a part of a circle.

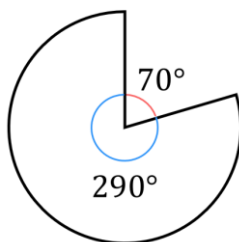
The **interior** angle of a sector is the angle **inside** the sector.

The interior angle and exterior angle must sum to 360°.

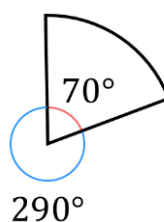


Identify the interior angle in these sectors

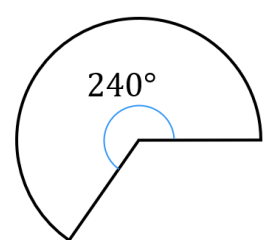
a



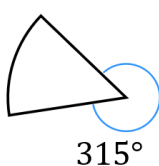
b



c



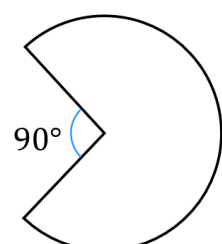
d



e



f



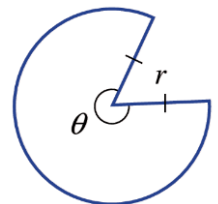
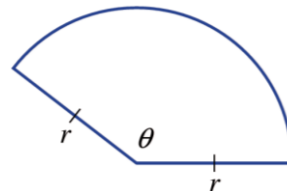
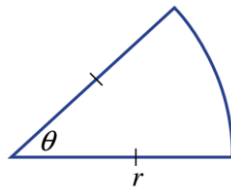
i Area of a Sector

A **sector** is a part of a circle.

The area of a sector is a fraction of the whole circle.

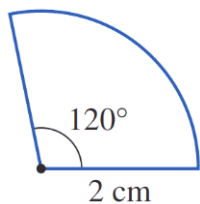
interior angle
↓
 $A = \frac{\theta}{360} \times \pi r^2$

fraction of circle area of whole circle



Example Calculate area of a sector

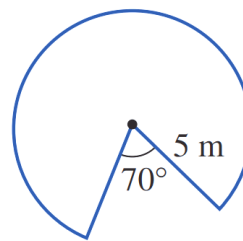
Calculate the area of these sectors. Answer to 2 decimal places.



$$A = \frac{\theta}{360} \pi r^2$$

$$= \frac{120}{360} \pi (2)^2$$

$$\approx 4.19 \text{ cm}^2$$



$$A = \frac{\theta}{360} \pi r^2$$

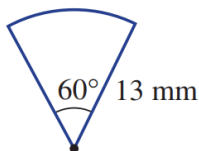
$$= \frac{290}{360} \pi (5)^2$$

$$\approx 63.27 \text{ m}^2$$

Guided Practice

Calculate the area of these sectors to 2 decimal places.

a



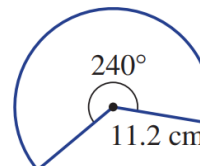
$$A = \frac{\theta}{360} \pi r^2$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

88.49 mm²

b



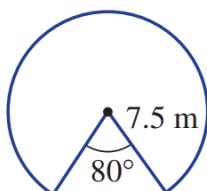
$$A = \frac{\theta}{360} \pi r^2$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

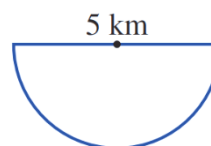
262.72 cm²

c



137.44 m²

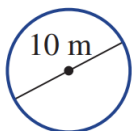
d



= 9.82 km²

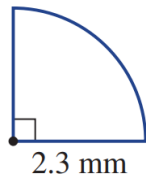
1 Identify the radius of these shapes.

a



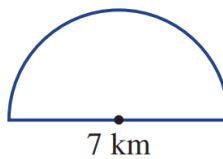
5 m

b



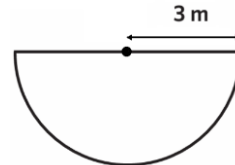
2.3 mm

c



3.5 km

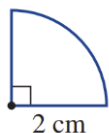
d



3 m

2 Calculate the area of these parts of circles. Round to 2 d.p.

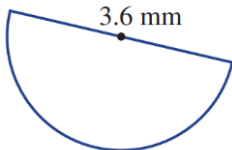
a



$$\begin{aligned} A &= \frac{1}{4}\pi r^2 \\ &= \frac{1}{4}\pi(\quad)^2 \\ &= \dots\dots\dots \end{aligned}$$

3.14 cm²

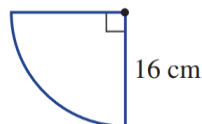
b



$$\begin{aligned} A &= \frac{1}{2}\pi r^2 \\ &= \frac{1}{2}\pi(\quad)^2 \\ &= \dots\dots\dots \end{aligned}$$

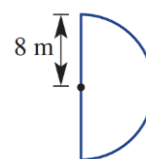
5.09 mm²

c



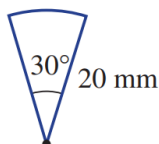
201.06 cm²

d



100.53 m²

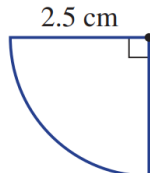
e



$$\begin{aligned} A &= \frac{\theta}{360} \times \pi r^2 \\ &= \frac{\quad}{360} \times \pi(\quad)^2 \\ &= \dots\dots\dots \end{aligned}$$

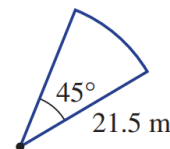
104.7 mm²

f



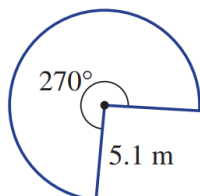
4.91 cm²

g



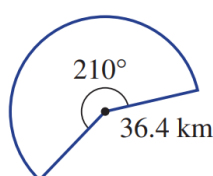
181.5 m²

h



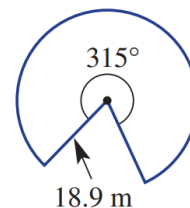
61.3 m²

i



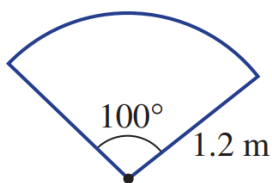
2428.1 km²

j



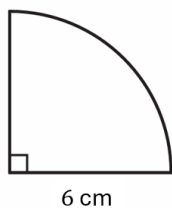
981.9 m²

- 3 A simple bus wiper blade wipes an area over 100° , as shown.
Find the area wiped by the blade, correct to two decimal places.

1.26 m²

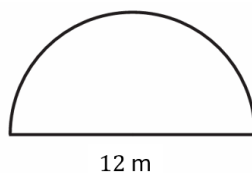
- 4 Calculate the area of each shape. Give your answers in terms of π .
The first has been done for you.

a

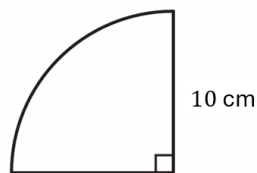


$$\begin{aligned} A &= \frac{\pi(6)^2}{4} \\ &= \frac{36\pi}{4} \\ &= 9\pi \text{ cm}^2 \end{aligned}$$

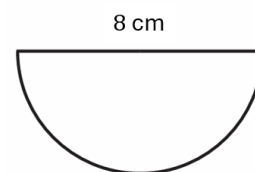
b

 $18\pi \text{ m}^2$

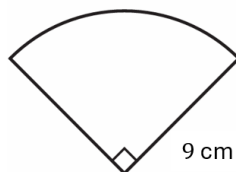
c

 $25\pi \text{ cm}^2$

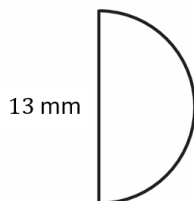
d

 $8\pi \text{ cm}^2$

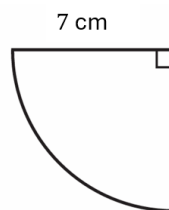
e

 $\frac{81}{4}\pi \text{ cm}^2$

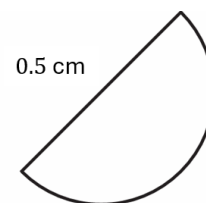
f

 $\frac{169}{8}\pi \text{ mm}^2$

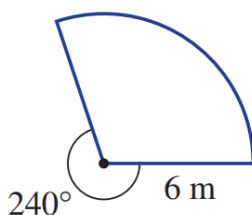
g

 $\frac{49}{4}\pi \text{ cm}^2$

h

 $\frac{\pi}{32} \text{ cm}^2$

- 5 For the sector shown, Jerry calculated the fraction of the circle as $\frac{240}{360} = \frac{2}{3}$.



- a Explain why he is wrong.

The angle inside the sector is not 240°

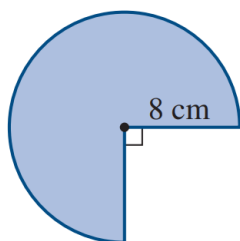
- b Calculate the area of the sector, round to 1 d.p.

37.7 m²

- 6 A shop sells pizza sectors. The two options are:
 15 cm radius pizza with an angle of 45° or 26 cm diameter pizza with an angle of 60° .
 Which piece gives the bigger area and by how much? Round the answer to two decimal places.

The 60° sector, by around 0.13 cm^2

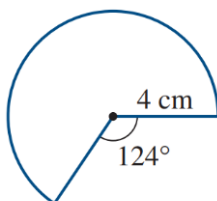
- 7 Calculate the area of the sector shown, answer in terms of π .



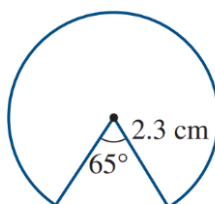
$48\pi \text{ cm}^2$

- 8 Calculate the area of these sectors. The angle shown is the exterior angle.
 Round your answers to 1 d.p.

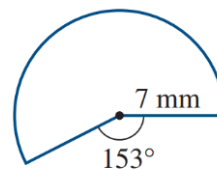
a



b



c

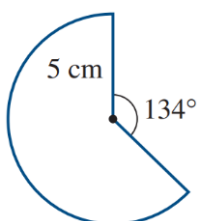


33.0 cm^2

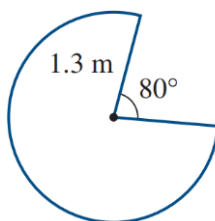
13.6 cm^2

88.5 mm^2

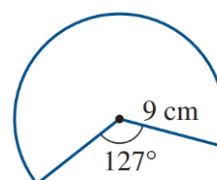
d



e



f



49.3 cm^2

4.1 m^2

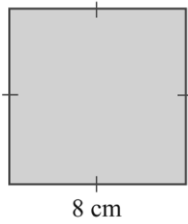
164.7 cm^2

Mixed Practice

FOUNDATION

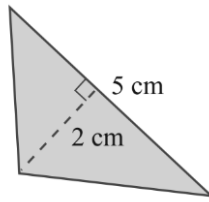
1 Calculate the area of these shapes. Answer to 2 d.p. where appropriate.

a



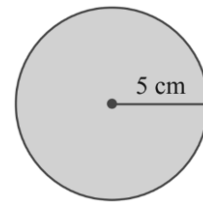
64 cm²

b



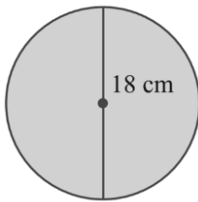
5 cm²

c



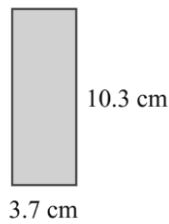
78.54 cm²

d



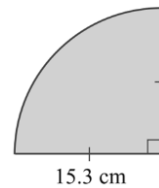
254.47 cm²

e



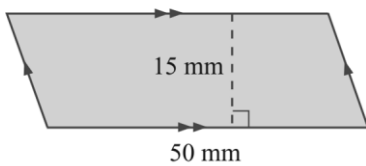
38.11 cm²

f



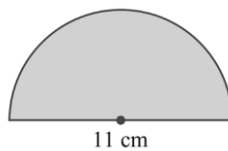
183.85 cm²

g



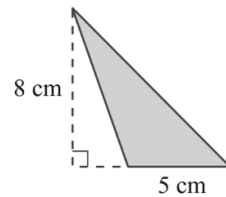
750 cm²

h



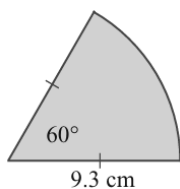
47.52 cm²

i



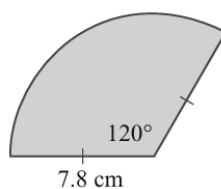
20 cm²

j



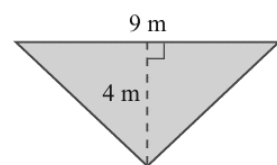
45.29 cm²

k



63.71 cm²

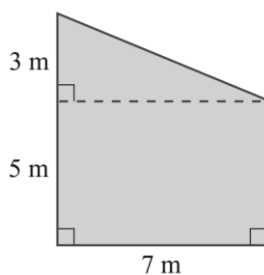
l



18 cm²

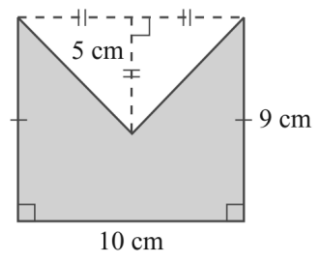
2 Calculate the area of these composite shapes.

a



45.5 m²

b



65 cm²

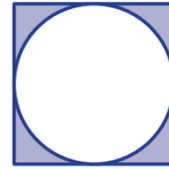
Composite Figures Involving Circles

i Area of Composite Figures

A **composite figure** is made up of multiple simple shapes.



A rectangle plus a quadrant



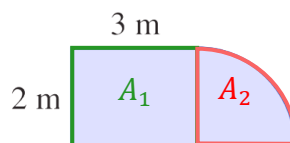
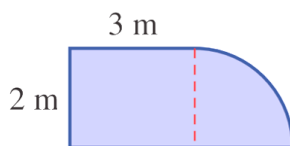
A square subtract a circle

To find the area of composite figures, add or subtract the areas of simple shapes.

1. Split the composite figure into simpler shapes.
2. Calculate the area of each simple shape.
3. Add or subtract the areas.

Example Calculate area of composite shapes involving circles

Calculate the shaded area to two decimal places.



$$\begin{aligned} A_1 &= 2 \times 3 \\ &= 6 \text{ m}^2 \end{aligned}$$

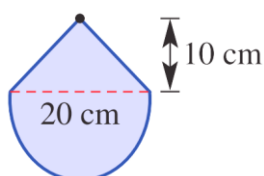
$$\begin{aligned} A_2 &= \frac{1}{4} \pi (2)^2 \\ &= 3.14 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} A_{\text{Total}} &= A_1 + A_2 \\ &= 6 + 3.14 \\ &= 9.14 \text{ m}^2 \end{aligned}$$

Guided Practice

Calculate the shaded area. Round your answer to 2 decimal places.

a



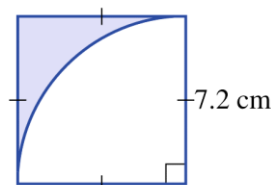
$$A_1 =$$

$$A_2 =$$

$$A_{\text{Total}} =$$

$$257.08 \text{ cm}^2$$

b



$$A_1 =$$

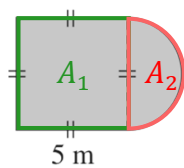
$$A_2 =$$

$$A_{\text{Total}} =$$

$$11.12 \text{ cm}^2$$

1 Calculate the areas of these composite shapes using addition. Answer to 2 d.p.

a



$$A_1 = \dots\dots\dots$$

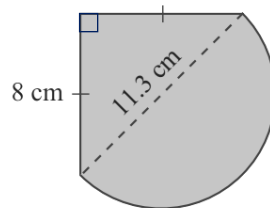
$$A_2 = \dots\dots\dots$$

$$A_{\text{Total}} = A_1 + A_2$$

$$= \dots\dots\dots$$

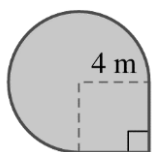
34.82 m²

b

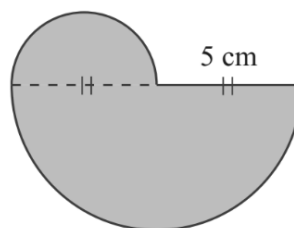


82.14 cm²

c



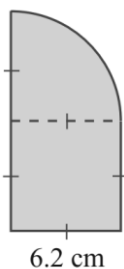
d



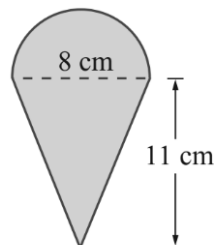
53.70 m²

49.09 cm²

e



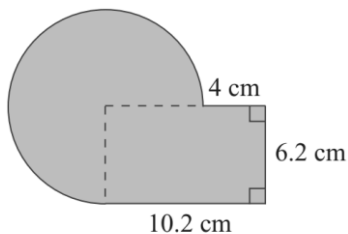
f



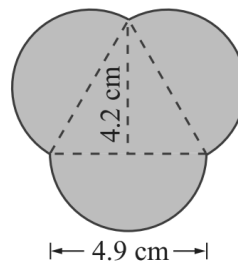
68.63 cm²

69.13 cm²

g



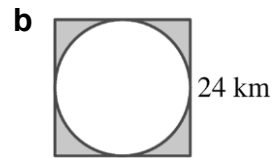
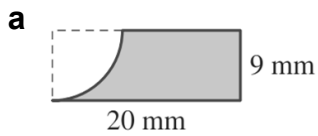
h



153.81 cm²

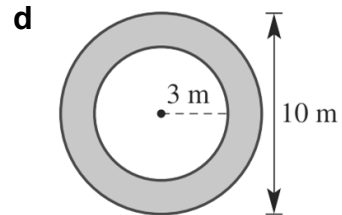
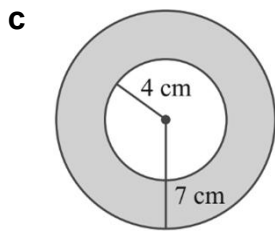
38.58 cm²

2 Calculate the areas of these composite shapes using subtraction.



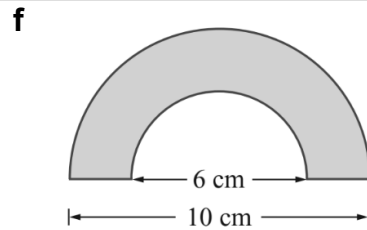
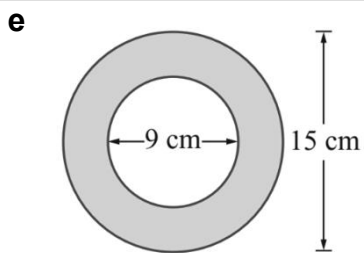
116.38 mm²

123.61 km²



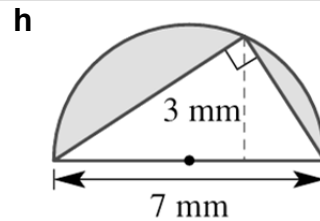
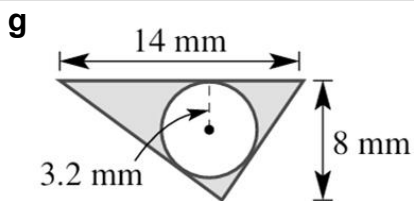
103.67 cm²

50.27 m²



113.09 cm²

25.13 cm²

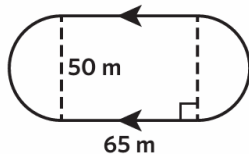


23.83 mm²

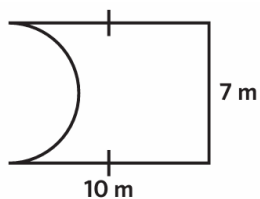
8.74 mm²

3 Calculate the area of these shapes. Answer to 2 decimal places.

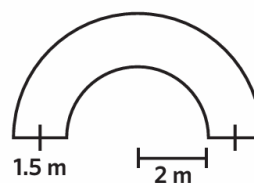
a

5213.50 m²

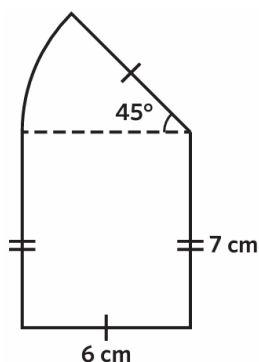
b

50.76 m²

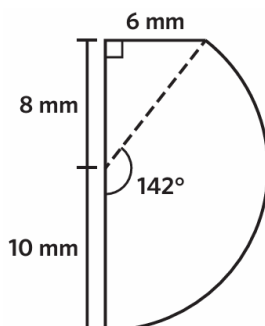
c

12.96 m²

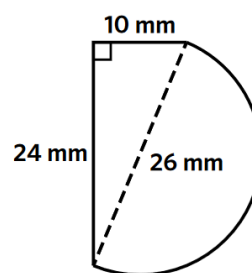
d

56.14 cm²

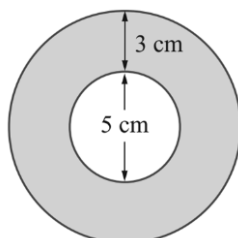
e

147.92 mm²

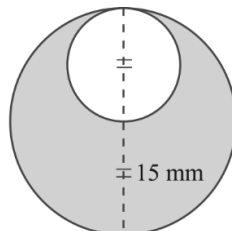
f

385.46 mm²

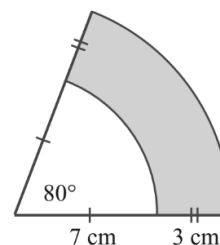
g

75.40 cm²

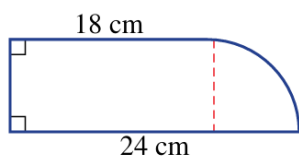
h

530.14 mm²

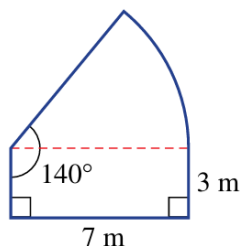
i

35.60 cm²

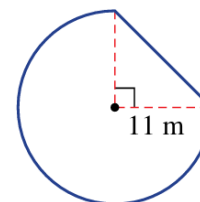
j

136.27 cm²

k

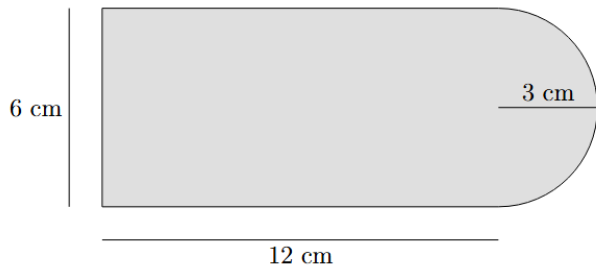
42.38 m²

l

345.60 m²

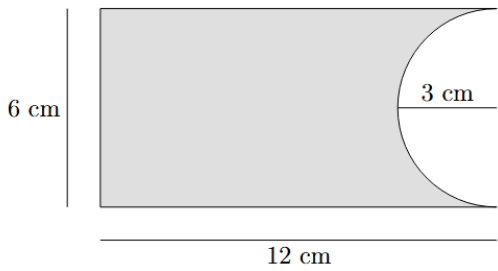
4 Calculate the shaded area of these shapes.

a



86.14 cm²

b



57.86 cm²

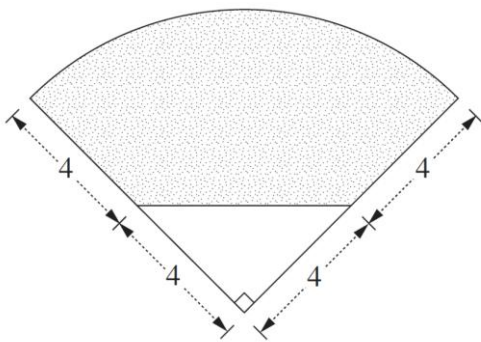
Add your areas together. What do you get?

$86.14 + 57.86 = 144$

This is double the rectangle area.

5 2009 HSC Standard 2 Band 4

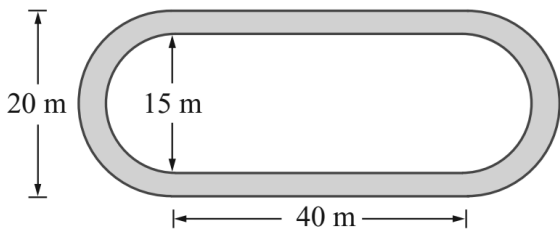
What is the area of the shaded part of this shape, correct to the nearest square centimetre?



NOT TO
SCALE
All measurements
are in centimetres.

$$\frac{1}{4}\pi(8)^2 - \frac{16}{2}$$
$$= 42.27 \text{ cm}^2$$

6 The racetrack shown needs to be resurfaced.
Calculate the area to be resurfaced to 2 decimal places.

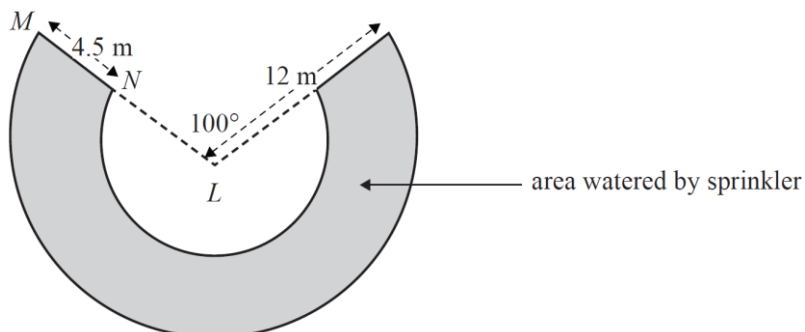


The racetrack is an oval made up of a rectangle and two semicircles.

Area is outer oval – inner oval.

$$A = (40 \times 20 + \pi(10)^2) - (40 \times 15 + \pi(7.5)^2)$$
$$337.44 \text{ m}^2$$

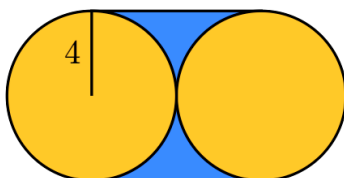
- 7 A golf course has a sprinkler system that waters the grass in the shape of a sector, as shown. A sprinkler is positioned at point L and can turn through an angle of 100° . The section of grass that is watered is 4.5 m wide at all points. Water can reach a maximum of 12 m from the sprinkler at L . What is the area of grass that this sprinkler will water? Round your answer to the nearest square metre.



$$A = \frac{260}{360}\pi(12)^2 - \frac{260}{360}\pi(7.5)^2$$

$$= 199 \text{ m}^2$$

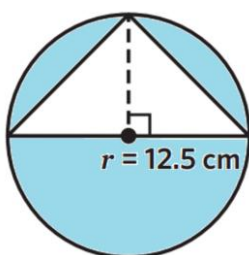
- 8 Find the area of the dark region to 2 d.p.



Area of dark region = Square – circle

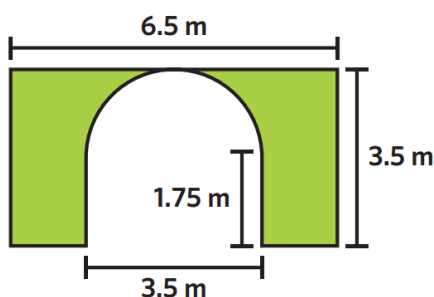
$$8^2 - \pi(4)^2 = 13.73 \text{ units}^2$$

- 9 A triangle is inscribed in a circle, as shown. Calculate the shaded area to 2 d.p.



$$\pi(12.5)^2 - \frac{1}{2}(25)(12.5) = 334.62 \text{ cm}^2$$

- 10 Tui has a wall that has a height of 3.5 m and width of 6.5 m. She has cut an arch into the wall that consists of a semicircle with a diameter of 3.5 m and a rectangle that is 1.75 m high. Calculate the area of the wall she has left, correct to 2 decimal places.



$$3.5 \times 6.5 - (3.5 \times 1.75) - \frac{1}{2}\pi(1.75)^2 = 11.81 \text{ m}^2$$

Finding the Radius Given the Area

Review

- ☐ Solve equations involving π , fractions, square roots.

Solve for r . Show the answer as a surd and to 1 d.p.

c $36 = \pi r^2$

d $50 = \pi r^2$

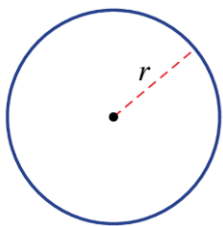
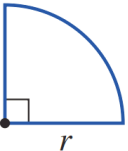
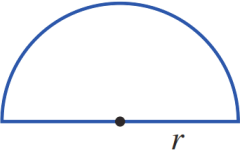
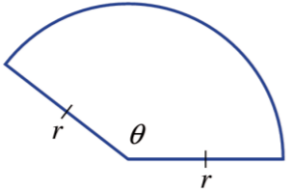
e $15 = \frac{\pi r^2}{2}$

f $8 = \frac{\pi r^2}{4}$

g $10 = \frac{50}{360} \times \pi r^2$

h $15 = \frac{100}{360} \times \pi r^2$

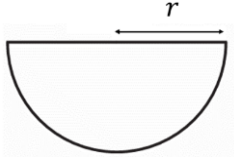
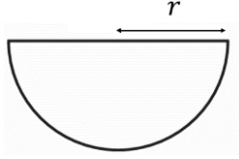
i Area of Circles

Circle	Quadrant	Semicircle	Sector
$A = \pi r^2$ 	$A = \frac{\pi r^2}{4}$ 	$A = \frac{\pi r^2}{2}$ 	$A = \frac{\theta}{360} \pi r^2$ 

Steps for using any formula:

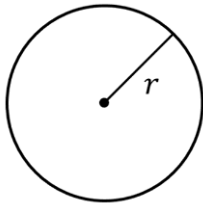
1. Write the formula.
2. Substitute given values.
3. Solve for the unknown value using inverse operations.

Example Solve for the radius, given the area

Example	Your Turn
The area of a semicircle is $A = 25 \text{ m}^2$ Find the radius to 3 d.p.  1. Write the formula $A = \frac{\pi r^2}{2}$ 2. Substitute $25 = \frac{\pi r^2}{2}$ 3. Solve $\begin{aligned} \times 2 \quad \times 2 \\ 50 &= \pi r^2 \\ \div \pi \quad \div \pi \\ \frac{50}{\pi} &= r^2 \\ \sqrt{\quad} \quad \sqrt{\quad} \\ \sqrt{\frac{50}{\pi}} &= r \\ 3.989 \text{ cm} &= r \end{aligned}$	The area of a semicircle is $A = 20 \text{ m}^2$ Find the radius to 3 d.p.  1. 2. 3. 3.568 m

 Guided Practice

- a** The area of a circle is 75 cm^2
Find the radius to 3 d.p.



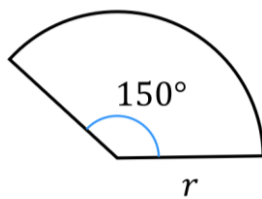
4.886 cm

- b** The area of a quadrant is $A = 13.5 \text{ m}^2$
Find the radius to 3 d.p.



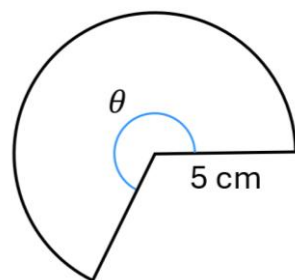
4.146 m

- c** The area of this sector is 30 m^2
Find the radius to 3 d.p.



4.787 m

- d** The area of this sector is 60 cm^2
Find the interior angle to the nearest degree.



275°

1 Calculate the radius of each circle, rounding your answer to 1 d.p.

a $A = 70 \text{ cm}^2$

$$A = \pi r^2$$

$$70 = \pi r^2$$

$$\frac{70}{\pi} = r^2$$

$$\sqrt{\frac{70}{\pi}} = r$$

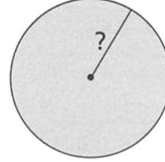
..... = r



4.7 cm

b $A = 98 \text{ cm}^2$

$$A = \pi r^2$$



5.6 cm

c $A = 0.45 \text{ m}^2$

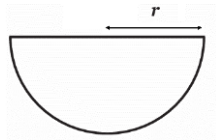


0.4 m

2 Calculate the radius of each semicircle or quadrant, rounding your answer to 2 d.p.

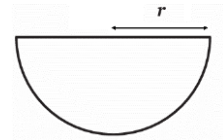
a $A = 15.5 \text{ mm}^2$

$$A = \frac{\pi r^2}{2}$$



3.14 mm

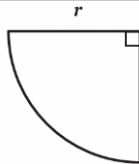
b $A = 14.8 \text{ cm}^2$



3.07 cm

c $A = 6 \text{ mm}^2$

$$A = \frac{\pi r^2}{4}$$



2.76 mm

d $A = 3.8 \text{ cm}^2$

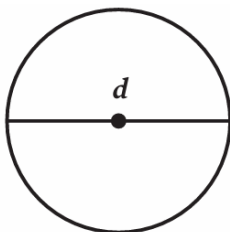


2.20 cm

3 Find the *diameter* for these areas, rounding your answer to 1 d.p.

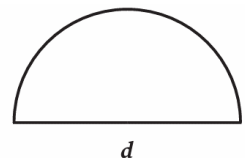
(Find the radius, then multiply by 2)

a $A = 12 \text{ cm}^2$



3.9 cm

b $A = 14 \text{ mm}^2$



6.0 mm

4 Read the example for when the area is given in terms of π and answer the questions.

a The area of a circle is $25\pi \text{ cm}^2$.
Calculate the radius.

$$\begin{aligned} A &= \pi r^2 \\ 25\pi &= \pi r^2 \\ \div \pi \quad \div \pi & \\ 25 &= r^2 \\ \sqrt{\quad} \quad \sqrt{\quad} & \\ 5 &= r \end{aligned}$$

The radius is 5 cm.

b The area of a circle is $9\pi \text{ cm}^2$.
Calculate the radius.

$$\begin{aligned} A &= \pi r^2 \\ 9\pi &= \pi r^2 \\ 9 &= r^2 \\ \dots &= r \end{aligned}$$

c The area of a circle is $100\pi \text{ cm}^2$.
Calculate the radius.

d The area of a semicircle is $32\pi \text{ cm}^2$.
Calculate the radius.

e The area of a quadrant is $9\pi \text{ cm}^2$.
Calculate the radius

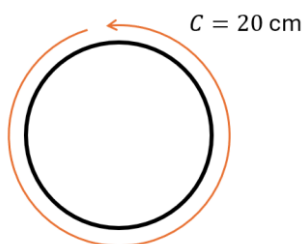
f The area of a quadrant is $23\pi \text{ cm}^2$.
Calculate the radius to 1 d.p.

8 cm

6 cm

10 cm

5 Ferdinand measures the circumference of his circular plate to be 20 cm.



a What is the radius of the plate?
(circumference is $C = 2\pi r$)

b What is the area that his plate covers?

3.183 cm

31.83 cm^2

6 Complete the table.

	Radius	Diameter	Area $A = \pi r^2$ (in terms of π)	Circumference $C = 2\pi r$ (in terms of π)
a	3	6	9π	6π
b	6	12	36π	12π
c	6	12	36π	12π
d	2	4	4π	4π
e	1	2	π	2π
f	4	8	16π	8π
g	8	16	64π	16π
h	$\frac{2}{3}$	$\frac{4}{3}$	$\frac{4\pi}{9}$	$\frac{4\pi}{3}$
i	$\frac{3}{10}$	$\frac{3}{5}$	$\frac{9\pi}{100}$	$\frac{3\pi}{5}$
j	$\frac{2}{5}$	$\frac{4}{5}$	$\frac{4\pi}{25}$	$\frac{4\pi}{5}$
k	$\frac{2}{25}$	$\frac{4}{25}$	$\frac{4\pi}{625}$	$\frac{4\pi}{25}$

7 NAPLAN A

The radius of a circle is 6.5 centimetres.

A square has the same area as this circle.

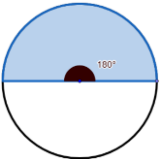
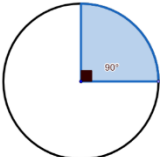
Calculate the side length of the square, in centimetres correct to one decimal place.

$$A = \pi(6.5)^2$$

$$l = \sqrt{\pi(6.5)^2}$$

$$11.52 \text{ cm}$$

8 Fill in the table.

	Sketch	Radius	Angle	Fraction of circle	Circumference	Arc Length	Area of circle	Area of sector
a		6	180°	$\frac{1}{2}$	12π	6π	36π	18π
b		6	90°	$\frac{1}{4}$	12π	3π	36π	9π
c		6	45°	$\frac{1}{8}$	12π	1.5π	36π	4.5π
d		6	270°	$\frac{3}{4}$	12π	9π	36π	27π
e		6	60°	$\frac{1}{6}$	12π	2π	36π	6π
f		6	120°	$\frac{1}{3}$	12π	4π	36π	12π
g		6	330°	$\frac{11}{12}$	12π	11π	36π	33π
h		12	330°	$\frac{11}{12}$	24π	22π	144π	132π
i		12	30°	$\frac{1}{12}$	24π	2π	144π	12π
j		12	90°	$\frac{1}{4}$	24π	6π	144π	36π

9

a Calculate the exact circumference of a circle with diameter 5 cm.

$5\pi \text{ cm}$

b Calculate the exact area of a circle with diameter 5 cm.

$\frac{25\pi}{4} \text{ cm}^2$

c The area of a circle is 48 cm^2 . Calculate the radius of the circle to 1 d.p.

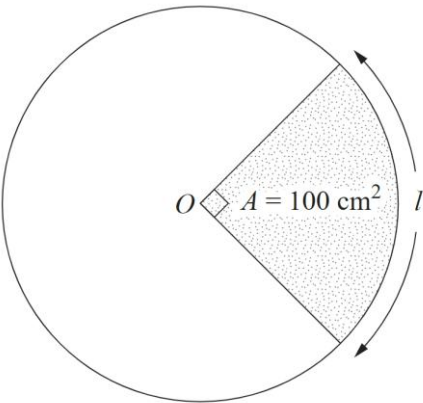
3.9 cm

d The circumference of a circle is 48 cm. Calculate the area to 1 d.p.

183.4 cm^2

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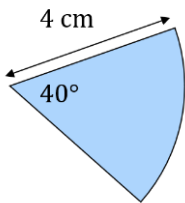
In the circle, the area of the quadrant is 100 cm^2 . What is the arc length l , correct to one decimal place?



$$100 = \frac{1}{4}\pi r^2$$
$$r = 11.283 \text{ cm}$$
$$l = \frac{1}{4}(2\pi r) = 17.7 \text{ cm}$$

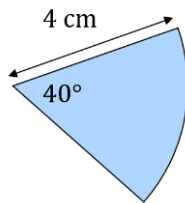
11 Consider the sector below. It is not drawn to scale.

a Calculate the area of the sector to 2 d.p.



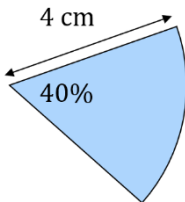
5.59 cm^2

b Calculate the perimeter to 2 d.p.



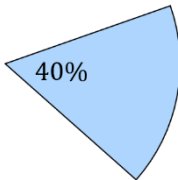
10.79 cm

c The sector is 40% of the circle. Calculate the area of the sector.



20.11 cm^2

d The sector is 40% of the circle and has an area of 200 cm^2 . Calculate the area of the whole circle.



$0.4 \times A = 200$
 $\therefore A = 500 \text{ cm}^2$

12 Find the unknown value in these sectors.

- a** The area is 16 cm^2 .
Find the radius to 2 d.p.



$$A = \frac{\theta}{360} \pi r^2$$

$$16 = \frac{50}{360} \pi r^2$$

$$16 = \frac{5}{36} \pi r^2$$

$$\times 36 \quad \times 36$$

$$576 = 5\pi r^2$$

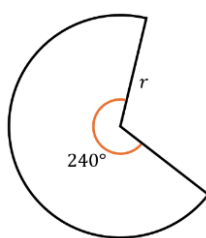
$$\div 5\pi \quad \div 5\pi$$

$$\frac{576}{5\pi} = r^2$$

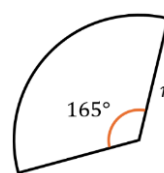
$$\sqrt{\quad} \quad \sqrt{\quad}$$

$$6.06 \text{ cm} = r$$

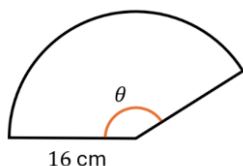
- b** The area is 105.6 cm^2 .
Find the radius to 1 d.p.



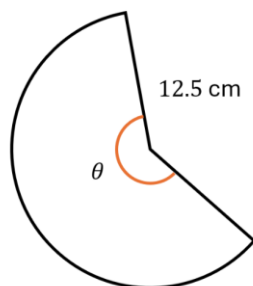
- c** $A = 323 \text{ cm}^2$
Find diameter to 2 d.p.



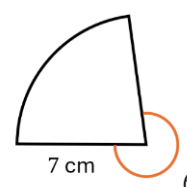
- d** The area is 335.1 cm^2 .
Find the angle.



- e** The area is 272.7 cm^2 .
Find the angle.



- f** The area is 32.07 cm^2 .
Find the outside angle.



150°

200°

285°

Challenge Exercise

- 1 Anastasia sells pizzas for \$24. Her pizzas have a radius of 18 cm. She is creating a new pizza size that has a radius of 21 cm. How much should she sell the new size so her customers pay the same amount per cm²?

$$\text{Current \$/cm}^2: \frac{\$24}{\pi(18)^2} = \$0.02357851009/\text{cm}^2$$

$$\text{New area: } \pi(21)^2 = 1385.44236 \text{ cm}^2$$

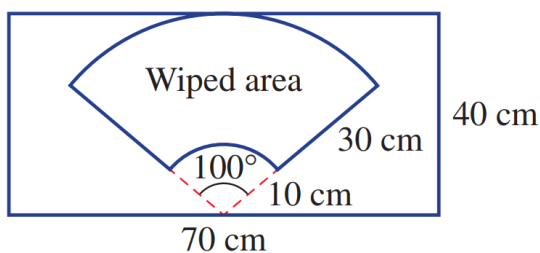
$$\begin{aligned} \text{New price} &= 1385.44236 \text{ cm}^2 \times \$0.02357851009/\text{cm}^2 \\ &= \$32.67 \end{aligned}$$

Alternate method

$$\text{new pizza is } \left(\frac{21}{18}\right)^2 \text{ as large.}$$

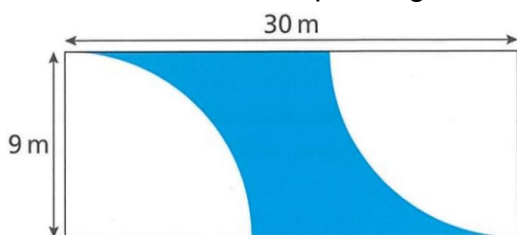
$$\$24 \times \left(\frac{21}{18}\right)^2 = \$32.67$$

- 2 Determine the percentage of the windscreen covered by the wiper to 2 d.p.



$$\frac{\frac{100}{360}\pi(40^2) - \frac{100}{360}\pi(10)^2}{40 \times 70} \times 100 = 46.75\%$$

- 3 The floor in an art gallery is to be painted. The diagram shows the region to be painted. A tin of paint covers 6.5 m². Each tin of paint costs \$12.95. Determine the cost of painting the floor.

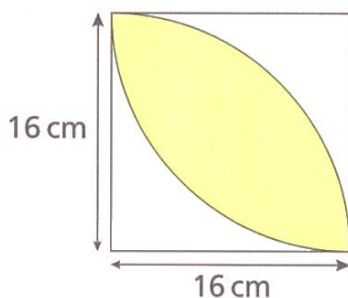


$$\text{Area} = 30 \times 9 - 2 \times \left(\frac{1}{4}\pi(9)^2\right) = 142.765 \text{ m}^2$$

$$\text{Tins of paint needed} = 142.765 \div 6.5 \approx 22 \text{ tins of paint needed}$$

$$22 \times 12.95 = \$284.90$$

- 4 Work out the area of the shaded region.
(Hint: use the area of a triangle!)

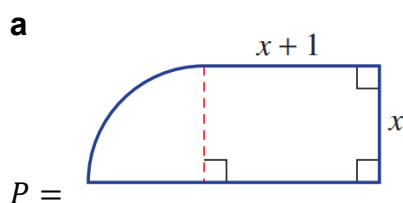


Consider a quadrant. The shaded segment is:

$$\text{Shaded segment} = \frac{1}{4}\pi(16)^2 - \frac{1}{2}(16)(16) = 73.06 \text{ cm}^2$$

There are two identical segments, so 146.12 cm^2

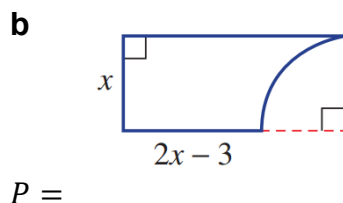
- 5 Find expressions, in simplest form, for the perimeter (P) and area (A) of these shapes.
Answer in terms of π ; for example, 4π . Do not use decimals.



$A =$

$$4x + 2 + \frac{\pi x}{2}$$

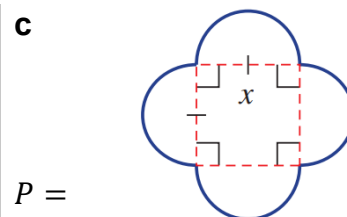
$$x^2 + x + \frac{\pi x^2}{4}$$



$A =$

$$6x - 6 + \frac{\pi x}{2}$$

$$2x^2 - 3x - \frac{\pi x^2}{4}$$

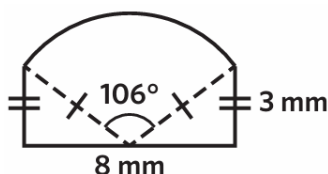


$A =$

$$4\pi x$$

$$x^2 + 2\pi x^2$$

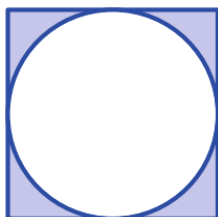
- 6 Find the area of the composite shape to 2 d.p.



Using right-angled triangle, $r = \sqrt{4^2 + 3^2} = 5$

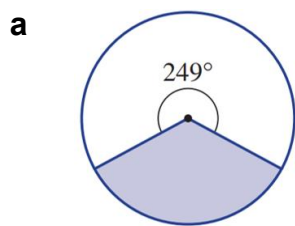
$$A = 2\left(\frac{1}{2} \times 3 \times 4\right) + \frac{106}{360}\pi(5)^2 = 35.13 \text{ mm}^2$$

- 7 What fraction of the square is shaded?

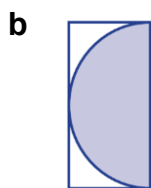


$$\frac{(2r)^2 - \pi r^2}{(2r)^2} = \frac{4r^2 - \pi r^2}{4r^2} = \frac{4 - \pi}{4}$$

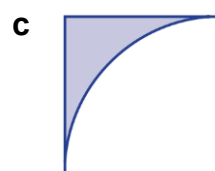
- 8 What percentage of the total area is occupied by the shaded region in these diagrams?
Round the answer to one decimal place.



$$\frac{111}{360} \times 100 = 30.8\%$$

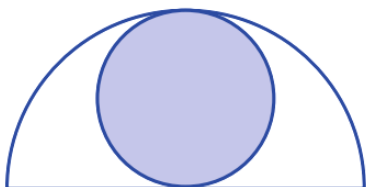


$$\begin{aligned} & \frac{\frac{1}{2}\pi r^2}{2r^2} \times 100 \\ &= \frac{\frac{1}{2}\pi}{2} \times 100 \\ &= 78.5\% \end{aligned}$$



$$\begin{aligned} & \frac{r^2 - \frac{1}{4}\pi r^2}{r^2} \times 100 \\ &= \frac{r^2 \left(1 - \frac{1}{4}\pi\right)}{r^2} \times 100 \\ &= 21.5\% \end{aligned}$$

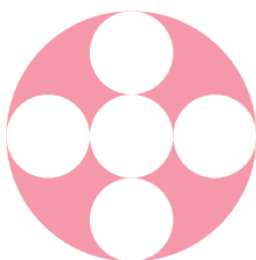
- 9 What percentage of the large semicircle is shaded?



Let radius of small circle be r

$$\begin{aligned} & \frac{\pi r^2}{\frac{1}{2}\pi(2r)^2} \times 100 \\ &= \frac{\pi r^2}{2\pi r^2} \times 100 \\ &= 50\% \end{aligned}$$

- 10 What percentage of the circle is shaded? Answer to 2 d.p.



Let radius of small circle be r

$$\begin{aligned} & \frac{\pi(3r)^2 - 4\pi r^2}{\pi(3r)^2} \times 100 \\ &= \frac{5\pi r^2}{9\pi r^2} \times 100 \\ &= 55.56\% \end{aligned}$$

- 11** The diagram shows two circles with the same centre.
The radius of the outer circle is equal to the diameter of the inner circle.
If the region between the two circles is divided into six equal segments,
what fraction of the circle is shaded in the diagram?



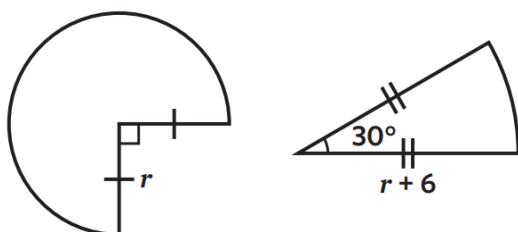
Let inner circle have radius r

$$\text{Area of ring: } \pi(2r)^2 - \pi(r)^2 = 3\pi r^2$$

Half the ring is shaded, so shaded area = $1.5\pi r^2$

$$\frac{1.5\pi r^2}{4\pi r^2} = \frac{3}{8}$$

- 13** The shapes in the diagram below have equal areas. Determine the value of r .



$$\frac{3}{4}\pi r^2 = \frac{1}{12}\pi(r+6)^2$$

$$\frac{3}{4}r^2 = \frac{1}{12}(r+6)^2$$

$$\frac{r^2}{(r+6)^2} = \frac{1}{9}$$

By trial and error, the radius is 3.

Alternate method using quadratic equations:

$$\frac{3}{4}r^2 = \frac{1}{12}(r^2 + 12r + 36)$$

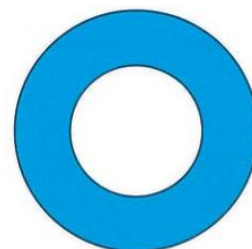
This simplifies to $-2r^2 + 3r + 9 = 0$

$$(-2r - 3)(r - 3) = 0$$

$r = 3$ or $-\frac{3}{2}$, but r must be positive.

$$r = 3$$

- 14** A shape is made up of two circles with the same centre.
The diameter of the large circle is 12 cm.
The area of the shaded region is 50 cm^2 .
Find the diameter of the smaller circle.



Area of the shaded region is $A = \pi R^2 - \pi r^2$

$$36\pi - \pi r^2 = 50$$

$$-\pi r^2 = 50 - 36\pi$$

$$r^2 = \frac{50 - 36\pi}{-\pi}$$

$$\therefore r = 4.48 \text{ cm}$$

$\therefore$ diameter of smaller circle is 8.96 cm